

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design Webpage using HTML5 & CSS3

2. AIM

To design and develop a personal portfolio webpage using **HTML5 and CSS3**, including navigation, home, about, skills, contact sections, styling, and hover effects..

3. OBJECTIVE

- The objectives of this experiment are:
 - To understand the basic structure of HTML5.
 - To use semantic HTML elements to create webpage sections.
 - To apply CSS3 properties for webpage design.
 - To create a navigation bar with links.
 - To design skill cards using CSS.
 - To create a contact form.
 - To implement hover effects on links and buttons.
 - To use Flexbox for arranging webpage elements.
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4. THEORY

HTML5 is used to create the structure and content of a webpage. Elements such as `<header>`, `<nav>`, `<section>`, `<form>`, and `<footer>` help organize the webpage.

CSS3 is used to style and design the webpage. Properties such as background, color, padding, margin, border-radius, box-shadow, and display: flex are used to improve the appearance and layout.

The `:hover` pseudo-class is used to change the appearance of links and buttons when the mouse pointer is placed over them..

The basic flow is:

HTML5 Structure → CSS3 Styling → Browser → Designed Webpage

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
 2. Define the HTML5 document structure.
 3. Add a `<header>` containing the portfolio title and navigation links.
 4. Create Home, About, Skills, and Contact sections.
 5. Add skill cards for HTML5, CSS3, and Python.
 6. Create a contact form with input fields and a textarea.
 7. Add CSS styles for colors, fonts, spacing, and layout.
 8. Use Flexbox to arrange the skill cards.
 9. Apply rounded corners and box shadows to the cards.
 10. Add hover effects to navigation links and buttons.
 11. Add a footer to the webpage.
 12. Save the file and open it in a web browser.
 13. Verify the webpage and its interactive effects.
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6. PROGRAM

Html&css.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Portfolio</title>
  <style>body { font-family:
Arial, sans-serif; background:
#f5f5f5; color:
#333;
}
header { background: #222;
color: white; padding: 10px
40px; display: flex; justifycontent:
space-between; alignitems: center;
} nav a { color:
white;
text-decoration: none; margin-left:
10px;
}
nav a:hover {
color: #00adb5;
}
.home { height: 200px;
background: #00adb5;
color: white; display: flex;
justify-content: center;
align-items: center; text-
align:
center;
}
.home h2 { font-size:
35px; margin-bottom:
15px;
}
.home p { font-size:
20px; margin-bottom:

```

```
25px;
} button
{
background: #222;
color: white; border:
none; padding: 12px
25px; borderradius:
5px; cursor:
pointer;
}
button:hover {
background: white; color:
#222;
}
```

```
.about { text-align: center;
padding:
30px 20px;
}
.about h2, .skills h2,
.contact h2 {
margin-bottom: 8px;
}
.skills { padding: 10px
20px; text-align: center;
}
.skill-container { display:
flex; justify-content:
center; gap:
25px;
}
.skill {
background: white; width:
250px; padding: 30px; border-
radius: 10px; boxshadow: 0 2px
10px #ccc;
}
.skill h3 { color: #00adb5;
margin-bottom:
5px;
```

```

}

.contact {  text-align: center;
padding:
10px 20px;
}
form {  width:
350px;  margin:
auto;
}
input, textarea {  width:
100%;  padding: 12px;
margin: 10px 0;  border:
1px solid #ccc;  border-
radius: 5px;
}
textarea {  height:
100px;  resize:
none; }</style>
</head>
<body>
  <header>
    <h1>My Portfolio</h1>
    <nav>
      <a href="#home">Home</a>
      <a href="#about">About</a>
      <a href="#skills">Skills</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>
  <section id="home" class="home">
    <div>
      <h2>Hello</h2>
      <p>Web Developer | Python Programmer | Student</p>
      <button>View </button>
    </div>
  </section>
  <section id="about" class="about">
    <h2>About Me</h2>

```

learning web development and programming.
creating websites and solving programming problems.

</p>

</section>

<section id="skills" class="skills">

<h2>My Skills</h2>

<div class="skill-container">

<div class="skill">

<h3>HTML5</h3>

<p>Creating webpage structure.</p>

</div>

<div class="skill">

<h3>CSS3</h3>

<p>Designing beautiful webpages.</p>

</div>

<div class="skill">

<h3>Python</h3>

<p>Solving programming problems.</p>

</div>

</div>

</section>

<section id="contact" class="contact">

<h2>Contact </h2>

<form>

<input type="text" placeholder="Your Name">

<input type="email" placeholder="Your Email">

<textarea placeholder="Your Message"></textarea>

<button type="submit">Send Message</button>

</form>

</section>

<footer>

<p>© 2026 My Portfolio | All Rights Reserved</p>

</footer>

</body>

</html>

7. CODE EXPLANATION

Element	Description
<header>	Creates the webpage header.
<nav>	Creates navigation links.
<section>	Divides the webpage into different sections.
.home	Styles the home section.
.about	Styles the About section.
.skills	Styles the skills section.
.skill-container	Uses Flexbox to arrange skill cards.
.skill	Creates individual skill cards.
.contact	Styles the contact section.
<form>	Creates the contact form.
<input>	Accepts user input information.
<textarea>	Accepts multi-line user text input.
button	Creates and styles buttons.
:hover	Adds mouse-over effects to elements.
box-shadow	Adds a shadow effect around skill cards.
border-radius	Creates rounded corners for elements.
display: flex	Arranges elements horizontally or vertically.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status:
TC-01	Open HTML file	Portfolio webpage should display	Webpage displayed	Pass
TC-02	Click Home link	Home section should appear	Home section displayed	Pass
TC-03	Click About/Skills link	Corresponding section should appear	Section displayed	Pass
TC-04	Enter name, email and message	Form should accept input	Input accepted	Pass
TC-05	Hover over navigation link	Link color should change	Hover effect displayed	Pass
TC-06	Hover over button	Button style should change	Hover effect displayed	Pass

9. OUTPUT

Output :

Hello

Web Developer | Python Programmer | Student

[View](#)

About Me

learning web development and programming. creating websites and solving programming problems.

My Skills

HTML5

Creating webpage structure.

CSS3

Designing beautiful webpages.

Python

Solving programming problems.

My Skills

HTML5

Creating webpage structure.

CSS3

Designing beautiful webpages.

Python

Solving programming problems.

Contact

[Send Message](#)

10. RESULT

Thus, a **personal portfolio webpage** was successfully designed using **HTML5 and CSS3** with navigation, sections, skill cards, contact form, styling, and hover effects. The expected output was obtained successfully.
